Mobile Application Development (MAP711S)

Introduction

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Introduction

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Introduction

Course Introduction

- Prerequisite: Programming II
- Course format:
  - Primarily face-to-face, otherwise online sessions via MS Teams
- Platforms used: E-learning, Moodle
- Consultation hours:
  - Find out from your course facilitator.
- Assessments:
  - Quizzes, Tests, Individual Exercises/Projects, and Group Project.

Introduction

Landscape

- percentage mobile operating system market share:

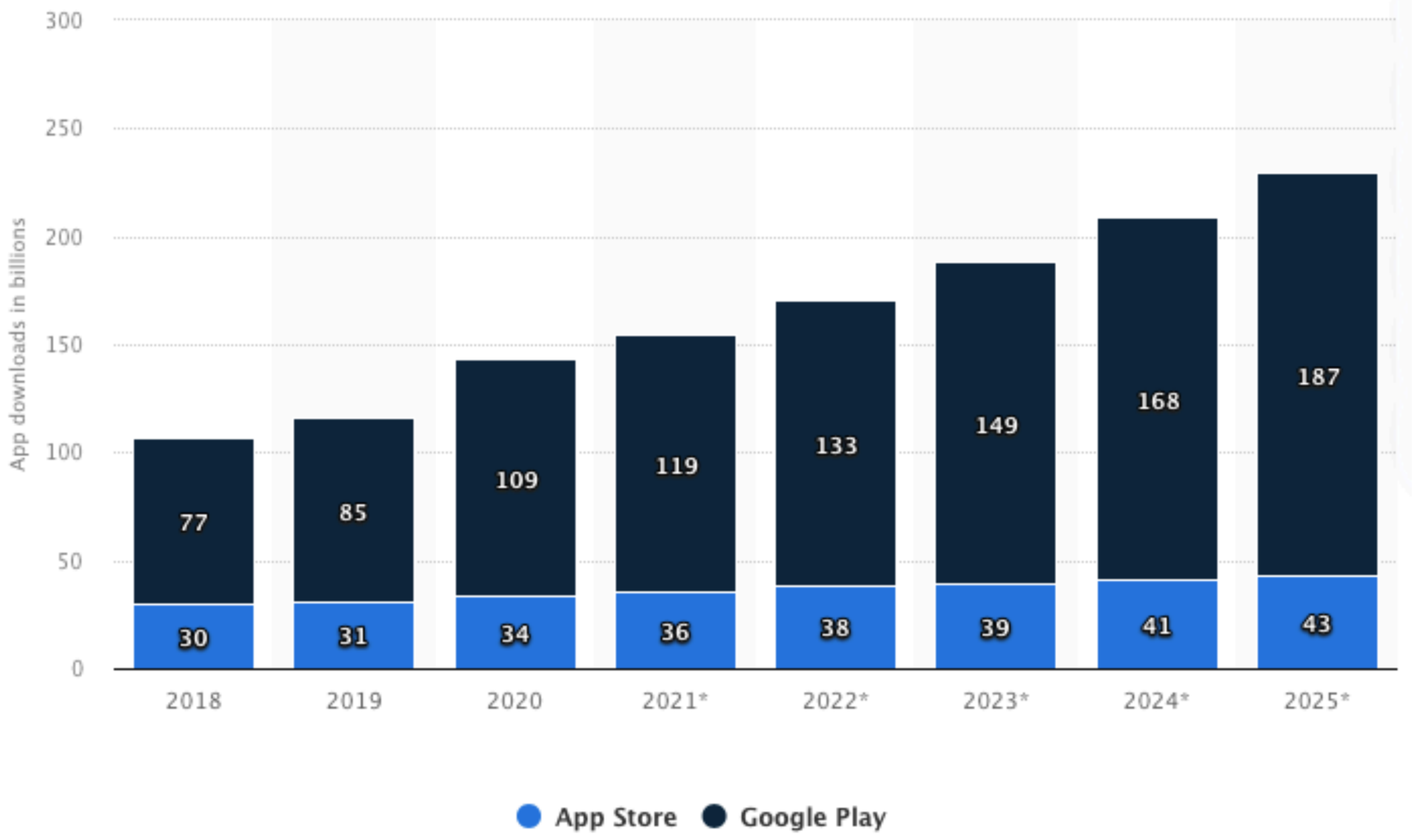
Region/	Android	iOS	Samsung	Other
Namibia	78.97	20.58	0.42	0.03
Global	70.36	29.26	0.25	0.03

- percentage mobile vendor market share:

Region/ Position	1st	2nd	3rd	4th	5th	6th
Namibia	Apple 57.20	Samsung 20.58	Huawei 6.24	Unknown 4.03	Intel 3.79	Xiaomi 2.04
Global	Apple 29.25	Samsung 20.30	Xiaomi 9.78	Unknown 8.91	Vivo 6.57	Oppo 6.27

Introduction

Mobile App Downloads Globally



(Statista, 2022)

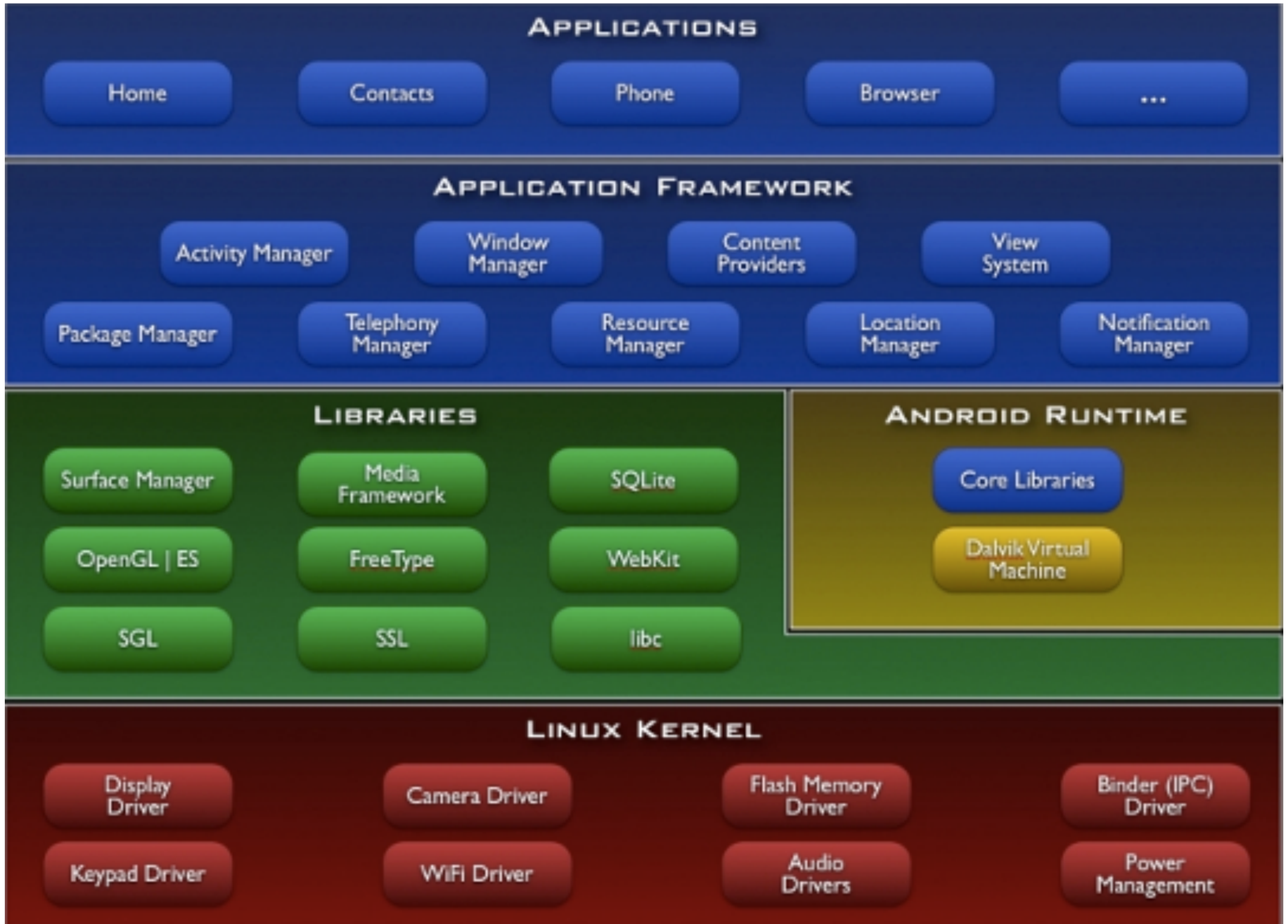
Introduction

Android Platform

- Android OS is a multi-user Linux system that treats each app as a different user.
- Each process has its own VM, thus each app's code runs in isolation from other apps.
- However, apps can share data through permissions or using the same ID.
- The Android system implements the principle of least privilege.
- Android apps written in Kotlin, Java, or C++.

Android Introduction

Android Architecture



Introduction

Android Studio

Android Introduction

SDK Packages

- SDK Platform - an Android version required by for compilation during development;
- Platform tools - includes various tools including the adb tool;
- SDK Tools - includes Android emulator & ProGuard;
- Intel or ARM system images - required to run the emulator;
- Support Repository - for backward compatibility & services like Firebase & Google maps;
- Google Play Billing - allows integration of billing services to an app;
- Extras include sample apps & documentation.

Android Introduction

Android Open Source Project

- allows stakeholders to customise the Android stack to suite different specifications & drivers;
- make it possible to port devices & accessories to the Android platform. The hardware abstraction layer (HAL) is used as the standard link between hardware & the Android platform stack;
- ensure compatibility requirements are met through its compatibility test suit (CTS), resulting in quality and consistent user experience;

What could be the benefits & drawbacks of the AOSP?

Android Introduction

Android Devices

- Acer
- Motorola
- Lenovo
- HKC
- Sony Erickson
- Dell
- I-mobile
- ...

HTC
Samsung
Videocon
General Mobile
Alcatel (TLC Corporation)
High Screen
Huawei

LG
ZTE
Geeks Phone
Sony
Cherry Mobile
Garmin
i-mobile

Introduction

Android Phones & Tablets

	ANDROID PLATFORM VERSION	API LEVEL	CUMULATIVE DISTRIBUTION
5	Lollipop	21	
5.1	Lollipop	22	99.9%
6	Marshmallow	23	99.6%
7	Nougat	24	99.2%
7.1	Nougat	25	98.8%
8	Oreo	26	98.4%
8.1	Oreo	27	97.6%
9	Pie	28	95.3%
10	Q	29	90.8%
11	R	30	83.0%
12	S	31	69.3%
13	T	33	57.9%
14	U	34	44.0%
15	V	35	26.8%
16	B	36	7.5%

Nougat

User Interface

Multi-window Support
Notifications
Quick Settings Tile API
Custom Pointer API

Performance

Profile-guided JIT/AOT Compilation
Quick Path to App Install
Sustained Performance API
Frame Metrics API

Battery Life

Doze on the Go
Project Svelte: Background Optimizations
SurfaceView

Wireless & Connectivity

Data Saver
Number Blocking
Call Screening

Graphics

Vulkan API

System

Direct Boot
Multi-locale Support, More Languages
ICU4J APIs in Android
APK Signature Scheme v2

Android for Work

Work profile security challenge
Turn off work
Always on VPN
Customized provisioning

Accessiblity

Vision Settings on the Welcome screen

Security

Key Attestation
Network Security Config
Default Trusted Certificate Authority

VR

Platform support and optimizations for VR Mode

Printing Framework

Print service enhancements

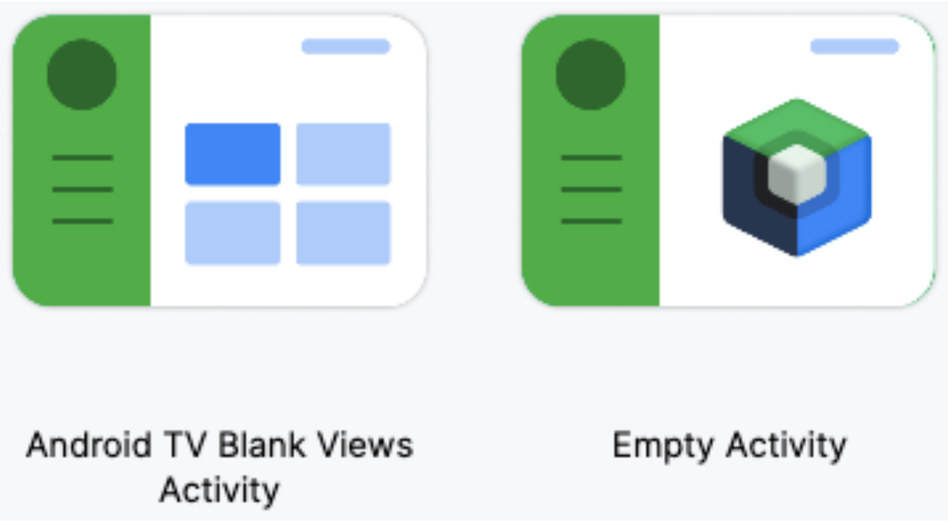
Introduction

Android Go Variant

- targets entry-level phones constrained in data, storage, memory, etc.
- minimum memory requirement is 512MB for Android 8.1 (API level 27).
- any Android app technically runs Go, but with significant built-in performance, network, and battery limitations.
- by default, Android Go disables features s.a., picture-in-picture support, System alert Window permission (draw over other apps), Split-screen or multi-screen window, Live wallpapers, Multi-display, VR mode, etc.
- as such, developing for Android Go requires special attention to resource usage.

Introduction

Android TV



- provides a big screen experience, usually at least 10 feet away.
- relies on a directional pad and a select button for navigation.
- also bear in mind that TVs are communal devices.

Introduction

Wear OS

Introduction

Android Auto

- optimise apps for drivers and cars.
- Apps should be simple, consistent, and free from distractions.
- The target is media, communication, and parked apps.

Introduction

Android XR

- creates immersive and augmented realities for users.
- targets special devices s.a. XR headsets, XR glasses, or AI glasses.
- development tools to consider include Jetpack XR SDK, Unity, Godot, Unreal Engine, or WebXR.
- Most Android apps are compatible with Android XR without additional development effort.
- it’s possible to spatialise existing layouts and enhance experiences with 3D models and immersive environments.

Introduction

Apple

- released iOS in 2007;
- leading tablet provider & also leads the smartwatch market;
- iOS for smartphones (iPhone & iPod touch), iPadOS for tablets, watchOS for smartwatches, & tvOS for televisions;
- Xcode used as the IDE;
- programming languages: Swift or Objective C;
- runs App Store for exclusive distribution of iOS apps;
- Enterprise Program allows distribution to a dedicated audience.

Introduction

Developing for iOS

- all forms of iOS take the same technical foundation of creating apps with varying elements to different form factors;
- macOS is needed to develop for iOS;
- an Apple ID & a yearly subscription of USD$99 to the Apple Developer Program is required before releasing an application to a beta testing group;
- SwiftUI was introduced in 2019 as a new way of developing user interfaces on all Apple platforms;
- Catalyst allows code sharing between iOS apps & its desktop environment;
- Develop for iOS: https://developer.apple.com/develop/

Introduction

Mobile Development: Opportunities

- Mobility:
  - easy to carry around.
- Convenience:
  - accessible all the time, always connected.
- Revenue:
  - developers can generate income.

Introduction

Mobile Development: Challenges

- Multiple Platforms:
  - phones, tablets, foldables, TV, car, XR
- Resource Limitations
  - memory, computation, and wireless connectivity.

Introduction

Connectivity

- a significant number of users still using 2G.

How can user experience be improved?

  - Optimise for low-speed connections
  - Provide offline capabilities
  - Queuing requests
  - Handling images for optimal performance

Introduction

Expensive Connections

- Namibia was ranked 5th most expensive in Africa (Statista, 2021).

How can user experience be improved?

- reduce app size.
- give configurable network options;
- be transparent on network costs.

Introduction

Device Limitation

- Android supports an increasing number of platforms and device specifications;

How best can user experience be improved?

  - support multiple screen sizes;
  - efficient memory management;
  - manage battery consumption.

Introduction

Software Concerns

- platform competition;
- fast-changing APIs;
- constantly changing user requirements;
- multiple form factors (not much of a concern nowadays);
- energy-aware software.

Mobile World

Competitive Mobile World

Do you remember any of these phones?

Introduction

Bracing Up for Competition

- Vendor:
  - partnerships, research, support, marketing, etc.
- Developer
  - Which platform to develop for? Market share, renumeration, etc.
  - Is the support for the platform sufficient? Tools, documentation, etc.
  - Is the learning curve good? Ease to learn, also skills already accrued may influence choice, personal development drive, etc.
  - collaboration, marketing, personal development, innovation, etc.

Introduction

Further Study

- Mobile OS Market Share: https://gs.statcounter.com/
- Android Developer:

  https://developers.google.com/speed/webp/docs/using
- Accessible Apps: https://medium.com/google-design/to-make-apps-accessible-make-them-compatible-with-different-devices-11298c6d3f06
- Transparency of data costs: https://medium.com/google-design/nurture-trust-through-cost-transparency-b61a5947d2fc

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- KaiOS: https://www.kaiostech.com
- Kotlin docs: https://kotlinlang.org/docs/home.html
- Android XR:

  https://developer.android.com/docs/quality-guidelines/android-xr
- Sailfish: https://sailfishos.org
- Samsung DeX: https://www.samsung.com/global/galaxy/apps/samsung-dex/
- Statista: https://www.statista.com/topics/779/mobile-internet/#dossierKeyfigures

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Thank You

Questions?

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